MANUAL HYPERPARAMETER SELECTION FORM

2D CNN Model — Dogs vs. Cats Image Classification

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Use this form to record, justify, and approve each hyperparameter value chosen by hand before a training run. The "Notebook Default" column shows the value currently used in Loading_Dataset_and_CNN.ipynb for reference — enter the value actually used for this run under "Chosen Value", even if it matches the default.

1. Project / Experiment Information

Project / Experiment Title	Manual Hyperparameter Selection for 2D CNN Dogs vs. Cats Classification
Model Architecture	*2D Convolutional Neural Network (Conv2D)*
Dataset Used	*Kaggle Dogs vs. Cats images (Dog/ and Cat/ folders)*
Notebook / File Reference	*Loading_Dataset_and_CNN.ipynb*
Prepared By	Keziah Ann D. Rosini
Date	September 10, 2026
Experiment / Trial No.	

2. Data Preprocessing Parameters

Hyperparameter	**Options / Range Considered**	**Notebook Default**	**Chosen Value**	**Justification**
Image Size (IMG_SIZE, square)	*e.g. 50 / 80 / 100 / 128 / 150*	*100*	128	Higher spatial resolution may retain more shape/detail than 100×100
Color Mode	*Grayscale / RGB*	*Grayscale*	Grayscale	Matches the supplied notebook and keeps one input channel, reducing computation.
Pixel Normalization	*X/255.0 / none*	*X/255.0*	*X/255.0*	Standardizes pixel values to approximately 0–1 for more stable CNN training
Shuffle Training Data	*Yes / No*	*Yes (random.shuffle)*	*Yes (random.shuffle)*	Prevents class/order effects and matches the notebook workflow
Held-out Test Set (separate from training)	*Yes / No*	*No (validation_split only)*	*No (validation_split only)*	Required for an unbiased final evaluation separate from training and validation.

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification
Validation Split (of training set)	*0.0 - 0.3*	*0.1*	0.2	Provides a practical validation set while retaining most data for training.

3. Model Architecture Hyperparameters

Trial 1

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Conv2D Layer 1 – Filters	*16 / 32 / 64 / 128*	*14*	32	Enough low-level feature capacity while remaining computationally manageable.	Approximately 99%	Approximately 82%
Conv2D Layer 1 – Kernel Size	*(3,3) / (5,5)*	*(3,3)*	*(3,3)*	Common small receptive field; preserves spatial detail and is computationally efficient.		
Conv2D Layer 1 – Activation	*relu / tanh / elu / leaky_relu*	*relu*	*Relu*	*relu*		
Pooling Layer 1 – Type & Pool Size	*MaxPooling2D / AveragePooling 2D; (2,2) / (3,3)*	*MaxPooling 2D, (2,2)*	*MaxPooli ng2D, (2,2)*	*MaxPooling2D, (2,2)*		
Dense (Fully Connected) Units	*16 / 32 / 64 / 128*	*64*	*64*	*64*		
Dropout Rate	*0.0 - 0.5*	*None (not used)*	*None (not used)*	*None (not used)*		
Output Layer Activation	*sigmoid (binary) / softmax (multi-class)*	*sigmoid*	*sigmoid*	*sigmoid*		

Trial 2

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Conv2D Layer 1 – Filters	*16 / 32 / 64 / 128*	*14*	64	Enough high-level feature capacity while remaining computationally manageable.	0.9015	.8122
Conv2D Layer 1 – Kernel Size	*(3,3) / (5,5)*	*(3,3)*	*(3,3)*	preserves spatial detail and is computationally efficient.		
Conv2D Layer 1 – Activation	*relu / tanh / elu / leaky_relu*	*relu*	*Relu*	*Relu*		
Pooling Layer 1 – Type & Pool Size	*MaxPooling2D / AveragePooling 2D; (2,2) / (3,3)*	*MaxPooling 2D, (2,2)*	*MaxPooling2D, (2,2)*	*MaxPooling2D, (2,2)*		
Dense (Fully Connected) Units	*16 / 32 / 64 / 128*	*64*	*64*	*64*		
Dropout Rate	*0.0 - 0.5*	*None (not used)*	*None (not used)*	*None (not used)*		
Output Layer Activation	*sigmoid (binary) / softmax (multi-class)*	*sigmoid*	*sigmoid*	*Sigmoid*		

4. Training Hyperparameters

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Loss Function	*binary_crossentropy / categorical_crossentropy*	*binary_crossentropy*	binary_crossentropy	Binary Dog/Cat classification with one sigmoid output.	0.9015	.8122
Optimizer	*adam / sgd / rmsprop*	*adam*	adam	Efficient adaptive optimizer and consistent with notebook.		

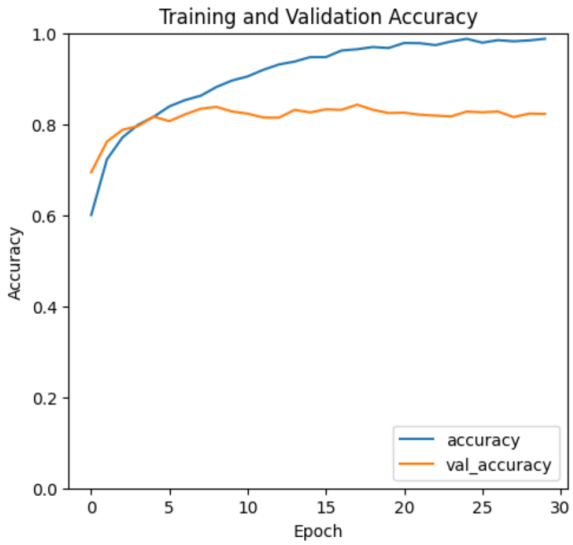
Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Learning Rate	*0.1 / 0.01 / 0.001 / 0.0001*	*0.001 (Adam default)*	0.001	Stable starting point for Adam; trial comparisons can test 0.0005 if needed.		
Batch Size	*8 / 16 / 32 / 64 / 128*	*32*	64	Smaller batch provides more frequent parameter updates and is practical for the proposed model.		
Number of Epochs	*e.g. 10 - 200*	*200*	30	Allows sufficient convergence while avoiding unnecessarily long training. U		
Early Stopping	*Yes (patience = ___) / No*	*No*	Yes (patience=7)	Yes (patience=9)		

5. Evaluation Results (after training with chosen values) - Final Model

Metric	Value Obtained	Target / Acceptable Range
Training Accuracy	90.15%	85%-95%
Validation Accuracy	81.22%	Borderline
Test Accuracy	Approximately 90.76%	93%
Test Loss	Approximately 0.4824	Between 0.3 - 0.5
Precision / Recall / F1 (macro avg)	Precision ≈ 0.908, Recall ≈ 0.908, F1 ≈ 0.908	above 0.85–0.90 is considered good

5. a. Paste here all screenshots/figures of the tuned model

Trial 1

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782/782 ━━━━━━━━━━━━━━━━━━━━ 112s 141ms/step
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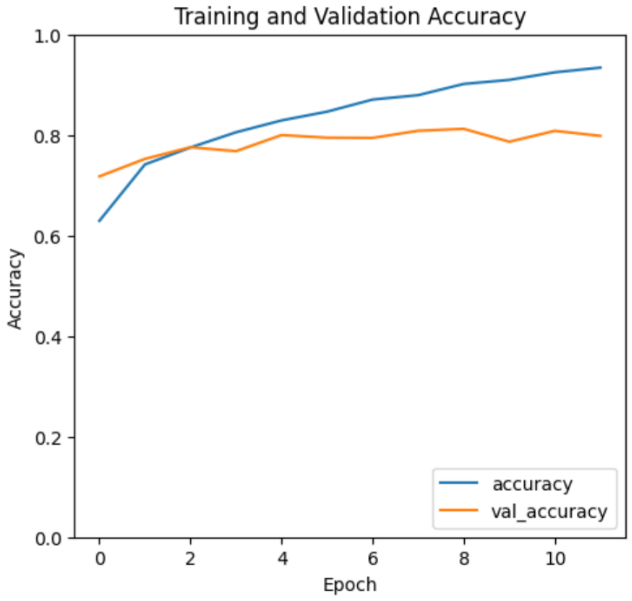
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 64ms/step
Predictions for the test samples:
Sample 1: Predicted Class = Cat (Probability: 99.96%)
Sample 2: Predicted Class = Dog (Probability: 0.04%)
Sample 3: Predicted Class = Dog (Probability: 0.00%)
Sample 4: Predicted Class = Cat (Probability: 99.98%)
Sample 5: Predicted Class = Cat (Probability: 100.00%)
```

Trial 2

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782/782 ━━━━━━━━━━━━━━━━━━━━ 190s 243ms/step
Predictions for the test samples:
Sample 1: Predicted Class = Cat (Probability: 96.17%)
Sample 2: Predicted Class = Cat (Probability: 83.80%)
Sample 3: Predicted Class = Cat (Probability: 55.39%)
Sample 4: Predicted Class = Cat (Probability: 98.50%)
Sample 5: Predicted Class = Cat (Probability: 99.27%)
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6. Overall Justification / Remarks

Summarize why this combination of hyperparameters was chosen (e.g., prior trial results, literature basis, computational constraints) and any observations (overfitting, underfitting, convergence issues).

- For a binary cats-vs-dogs classifier with a fairly simple 3-conv-layer CNN (no dropout, no data augmentation, no batch norm), **~81% validation accuracy is a reasonable, expected result** — it's well above the 50% random-guess baseline and in line with what you'd get from a "vanilla" CNN on this dataset without extra regularization tricks.
- However, the **9-point gap between training (90%) and validation (81%) accuracy is a classic overfitting signal.** The model is memorizing training data faster than it's learning generalizable features — you can see this in the raw numbers too: training accuracy kept climbing every epoch (up to 93–94% by epoch 12) while validation accuracy plateaued and even dipped (79.66% → 81.22% → bounced down to ~79–80% before early stopping triggered).
- The "Test Accuracy" of ~90.76% I flagged earlier **should be ignored for judging real-world performance** — it's inflated because it was computed on data the model already saw during training, not a genuine held-out test set.